

equipment. However, determining which contact initiated the trip can be challenging. In such situations, if the suspected and available dupli-

cate logic contacts of the circuit are wired to this 'fault finder,' the first faulty contact can be identified from the contact changeover time recorded in the laptop's Arduino IDE serial monitor. Thus, the fault can be easily found, analysed, and rectified in a brief time.

The prototype on a breadboard is shown in Fig. 1. The components required to build this device are given in the Bill of Materials table.

Circuit and working

BILL OF MATERIALS	
Components	Quantity
Resistors 10-kilo-ohm, ¼ watt (R0-R9)	10
STM32 F106C8T6 Blue Pill development board	1
USB adaptor	1

```

fault_finder_STM32_8IO.ino
11 byte contact1state;
12 byte lastcontact1state;
13 int VPin = PA0; //A0; (Arduino analog input)
14
15 int contact2 = PB4; //4; (Arduino input)
16 byte contact2state;
17 byte lastcontact2state;
18
19 int contact3 = PA15; //5; (Arduino input)
20 byte contact3state;
21 byte lastcontact3state;
22
23 int contact4 = PB6; //6; (Arduino input)
24 byte contact4state;
25 byte lastcontact4state;
26
27 int contact5 = PB8; //7; (Arduino input)
28 byte contact5state;
29 byte lastcontact5state;
30
31 int contact6 = PB9; //8; (Arduino input)
32 byte contact6state;
33 byte lastcontact6state;
34
35 int contact7 = PB10; //9; (Arduino input)
36 byte contact7state;
37 byte lastcontact7state;
38
39 int stopcontact = PB7; //2; (Arduino input)
40 byte stopcontactstate;
41 byte laststopcontactstate;
42
43
44 void entry1() {
45   Serial.print( "IN1 = ");
46   Serial.print(contact1state);
47   Serial.print("; Time(msec)= ");
48   Serial.println(millis());
49 }
50
51 void entry2() {
52   Serial.print( "IN2 = ");

```

Fig. 3: Code snippet

The circuit diagram of the fault finder is shown in Fig. 2. The STM32 board is connected with 8 digital inputs and an analogue input. It is powered by a 5V regulated power supply. All 8 digital I/O pins are chosen as 5V tolerant input pins so that the external contacts can be connected to a 5V power source. (For other regular input pins, it is necessary to connect only a 3.3V power supply.)

All the used digital I/O pins are pulled to ground through 10k resistors, and all the input external contacts are connected to 5V. Any logic state change is sensed by the board and logged/displayed on the Arduino IDE serial monitor.

The analogue input pin is kept at 1.15V by resistors R8 and R9. (These resistors can be changed or removed accord-

ing to the user's source of analogue input voltage.)

The program is written such that any deviation/abnormality in its value ($< 1.0V$ or $> 2.0V$) is used to log the value in the serial monitor. The user can change the set value in the program according to the field conditions.

The software for the fault finder is written using Arduino IDE version 1.8.7. The method of boot loading and uploading the Arduino program is very well explained in the DIY project 'LED scrolling display using STM32 controller' published in the EFY magazine September 2022 issue. Fig. 3 shows the snippet of the source code.

Following the procedure explained in the above-mentioned project, upload the source code 'fault_finder_STM32_8IO.ino' into the STM32 Blue Pill board.

Construction and testing

This device has only a few external resistors and connectors; the circuit can be assembled on a multipurpose PCB or breadboard and connected to the STM32 board. For easy and quick testing, assemble the circuit on a breadboard as per the circuit diagram (refer to Fig. 2).

All digital inputs I/P1 to I/P8 (CON-1 to -8) are connected

EFY Note

The source code of this project is available for free download at www.electronicsforu.com